

UserForm Interface Documentation

Overview

The Supply Chain Forecasting Tool includes two pre-built UserForm interfaces that provide a complete graphical user interface (GUI) for time series forecasting. **These forms are already included in the project** and do not need to be created programmatically.

UserForms Included

1. ForecastGUI - Single Component Forecasting Tool

File: ForecastGUI.frm

Purpose: Main interface for analyzing individual time series components using advanced forecasting methods.

Interface Layout

```

+-----+
| Time Series Forecasting Tool |
+-----+
| CSV File Path: |
| | [Browse...] |
| |
| Column Name: | Frequency: | Forecast Horizon: | |
| | | | |
| | 12 | | 12 |
| | | |
| Seasonal Type: |
| | Additive ▼ |
| |
| [Load Data] [Analyze] [View Charts] |
| [Export Results] [Close] |
|
| Status: Ready |
+-----+

```

Key Features

Input Controls:

- **CSV File Path** - TextBox for file path with Browse button
- **Column Name** - Specify which column contains time series data
- **Frequency** - Seasonal period (e.g., 12 for monthly data with yearly seasonality)
- **Forecast Horizon** - Number of periods to forecast ahead
- **Seasonal Type** - Dropdown: Additive or Multiplicative

Action Buttons:

- **Browse** - Open file dialog to select CSV file
- **Load Data** - Read and validate the CSV data
- **Analyze** - Run forecasting algorithms (SES and Holt-Winters)
- **View Charts** - Generate and display diagnostic charts
- **Export Results** - Save forecast results to CSV
- **Close** - Close the form

Status Bar:

- Real-time feedback on operations
- Color-coded: Blue (processing), Green (success), Red (error)

Functionality

1. Data Loading:

- Reads CSV files with custom column selection
- Validates data format and content
- Displays data point count

2. Analysis:

- Simple Exponential Smoothing (SES)
- Holt-Winters Exponential Smoothing (additive/multiplicative)
- Time series decomposition
- Automatic parameter optimization

3. Results:

- Creates separate sheets for each analysis type

- Generates diagnostic charts
- Calculates accuracy metrics (MAPE, MAE, RMSE)
- Provides 95% confidence intervals

2. BatchForecastGUI - Multi-Component Batch Processing

File: BatchForecastGUI.frm

Purpose: Advanced interface for processing 50-60+ components simultaneously with batch processing capabilities.

Interface Layout

Batch Forecasting Tool - Multi-Component Analysis

Processing Mode:
() Single Component (•) Batch Processing

CSV File Path:
[Browse...]

Column Name:
Frequency:
Forecast Horizon:

12
12

Seasonal Type:
Additive ▼

Data Format:
Wide Format (Period|Comp1|...) ▼

Quick Mode

Full Diagnostics

Components Loaded: 0 components

Progress: Processing 0 of 0

[Load Multi-Component Data]

[Batch Analyze All]

[View Charts]

[Export Results]

[Close]

Status: Ready - Select processing mode

Key Features

Mode Selection:

- ****Single Component**** - Process one time series (uses ForecastGUI logic)
- ****Batch Processing**** - Process multiple components simultaneously

Batch-Specific Controls:

- ****Data Format Dropdown****
- Wide Format: Period | Component1 | Component2 | Component3 ...
- Long Format: Component | Period | Value
- ****Processing Options****
- Quick Mode: Fast processing with summary statistics only
- Full Diagnostics: Detailed charts for top/bottom 10% components by MAPE

Progress Tracking:

- Real-time progress bar
- Current component being processed
- Count of completed components

Component Counter:

- Displays number of components loaded from CSV
- Updates after successful data load

Batch Processing Capabilities

1. Multi-Component Loading:

- Supports wide format (components as columns)
- Supports long format (stacked data)
- Automatically detects and counts components
- Validates data consistency

2. Batch Analysis:

- Processes all components in sequence
- Updates progress bar in real-time
- Shows current component name
- Handles errors gracefully (continues with next component)

3. Batch Results:

- Summary statistics for all components

- Accuracy metric rankings (MAPE, MAE, RMSE)
- Forecast values for all components
- Consolidated export format

4. Smart Diagnostics:

- Full diagnostics for best performers (lowest MAPE)
 - Full diagnostics for worst performers (highest MAPE)
 - Summary charts for overall performance
 - Component comparison visualizations
-

Implementation Details

Pre-Built Forms

Both UserForms are **already created** and included as .frm files in the VBA_Modules directory:

- `ForecastGUI.frm` - Contains the full UserForm with all controls and code
- `BatchForecastGUI.frm` - Contains the batch processing UserForm

AutoSetup Changes

The AutoSetup.bas module has been **updated** to:

- ✓ Skip programmatic UserForm creation (forms already exist)
- ✓ Verify that UserForms are present in the project
- ✓ Set up dashboard and other components only
- ✓ Faster setup process (2 steps instead of 3)

How to Use

1. **Open the Excel workbook** (must be .xlsx format)

2. **Import UserForms** (if not already imported):

- Open VBA Editor (Alt+F11)
- File → Import File

- Select `ForecastGUI.frm`
- Select `BatchForecastGUI.frm`

3. Run Setup:

```
`vba
AutoSetup.CreateCompleteApplication
`
```

4. Launch Forms:

```
`vba
' Single component
ForecastGUI.Show

' Batch processing
BatchForecastGUI.Show
`
```

Generating Screenshot Documentation

To generate a PDF with actual screenshots of the UserForms:

Prerequisites

1. **Windows OS** with Excel installed
2. **Python 3.7+** installed
3. **Required packages:**

```
`bash
pip install pywin32 pillow reportlab
`
```

Steps

1. Ensure Excel file exists:

- The script expects `TimeSeriesForecaster.xlsm` in the same directory
- Make sure UserForms are imported into the workbook

2. Run the capture script:

```
`bash  
cd ExcelForecaster  
python capture_userform_screenshots.py  
`
```

3. Output:

- Screenshots saved to `screenshots/` directory
- PDF generated: `UserForm\_Documentation.pdf`

What the Script Does

1. Opens Excel application (visible mode)
 2. Loads the workbook
 3. Displays each UserForm (modeless)
 4. Captures full-screen screenshot
 5. Closes UserForm
 6. Generates professional PDF with:
 - Title page
 - Each UserForm on separate page
 - Description of functionality
 - High-quality screenshots
-

Technical Specifications

ForecastGUI

- **Form Size**: 720 x 560 points
- **Controls**: 14 total
- 7 Labels
- 4 TextBoxes
- 1 ComboBox
- 6 CommandButtons
- **Code Lines**: ~640 lines of VBA
- **Key Functions**:
 - `UserForm\_Initialize()` - Setup defaults
 - `btnLoadData\_Click()` - CSV reader
 - `btnAnalyze\_Click()` - Run forecasting
 - `DisplayResults()` - Output to sheets
 - `CreateAllCharts()` - Visualization

BatchForecastGUI

- **Form Size**: 720 x 630 points (larger for batch features)
 - **Controls**: 20+ total
 - OptionButtons for mode selection
 - FrameControl for progress bar
 - Additional checkboxes for options
 - Dynamic labels for component count
 - **Code Lines**: ~387 lines of VBA
 - **Key Functions**:
 - `UpdateControlStates()` - Dynamic UI
 - `UpdateProgress()` - Real-time feedback
 - `optBatchComponent\_Click()` - Mode switching
 - Integration with `BatchProcessing.bas` module
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Integration with Main Modules

Both UserForms integrate seamlessly with the backend modules:

- **TimeSeriesAnalysis.bas** - Core forecasting algorithms

- **ChartUtilities.bas** - Chart generation
 - **MainModule.bas** - Data management and export
 - **BatchProcessing.bas** - Multi-component processing
-

Notes for Developers

Modifying UserForms

To modify the UserForms:

1. Open VBA Editor (Alt+F11)
2. Double-click the UserForm in Project Explorer
3. Enter Design Mode
4. Modify controls visually
5. Edit code in the Code window (F7)
6. Save workbook

Exporting UserForms

To share or backup UserForms:

1. Right-click UserForm in VBA Editor
2. Export File...
3. Save as .frm file
4. This also creates a .frx file (binary data for controls)

Version Control

Both .frm and .frx files should be committed to version control:

- .frm - Text file with VBA code and control properties
- .frx - Binary file with control images and data

Troubleshooting

UserForm Not Found Error

Problem: "UserForm 'ForecastGUI' not found"

Solution:

1. Check VBA Project Explorer for UserForm
2. If missing, import the .frm file
3. Ensure macro security allows VBA
4. Trust the workbook location

Controls Not Responding

Problem: Buttons or controls don't work

Solution:

1. Check if Design Mode is enabled (disable it)
2. Verify code is present (view code with F7)
3. Check for VBA errors in code
4. Ensure all required modules are imported

Screenshot Script Fails

Problem: Python script errors

Solution:

1. Run on Windows (not Linux/Mac)
2. Excel must be installed
3. Install all required packages
4. Close other Excel instances

5. Check file path is correct

Future Enhancements

Potential improvements for the UserForms:

- ■ Real-time chart preview in the form
 - ■ Drag-and-drop CSV file support
 - ■ Recent files dropdown
 - ■ Preset configurations
 - ■ Multi-language support
 - ■ Tooltips for all controls
 - ■ Form resizing support
 - ■ Dark mode theme
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